

VEXAR Fleet Analytics — Technical Report

Driver Behaviour & Vehicle Health Dashboards

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Tool Used: Microsoft Power BI Desktop

Dataset Period: 31 July 2026 – 06 August 2026

1. Problem Statement

VexarDrive Technologies operates a two-wheeler delivery and rider fleet across multiple hubs in Bengaluru. The objective of this assignment was to analyze one week of trip data — covering 30 drivers, 30 vehicles, and 450 trips — along with per-minute GPS and IMU sensor telemetry captured from drivers' mobile phones during each trip.

Using this dataset, two dashboards were required to be built:

1. **Driver Behaviour Dashboard** — to identify and score risky vs. safe driving patterns per driver.
2. **Vehicle Health Status Dashboard** — to identify vehicles showing signs of mechanical wear or irregular sensor signatures that may need maintenance.

For every metric and score shown across these dashboards, the reasoning and methodology is documented clearly below. All assumptions made during the process are also stated explicitly.

2. Dataset Structure & Relationships

The dataset consists of four tables joined as follows:

Telemetry [Trip_ID] → Trips [Trip_ID]

Trips [Driver_ID] → Drivers [Driver_ID]

Trips [Vehicle_ID] → Vehicles [Vehicle_ID]

Table	Grain	Key Columns
Drivers	One row per driver (Master Data)	Driver_ID, Age, Gender, License_Experience_Years, Home_Hub
Vehicles	One row per vehicle (Master Data)	Vehicle_ID, Make, Model, Manufacture_Year, Odometer_KM_Start_of_Week, Last_Service_Date
Trips	One row per trip (450 rows)	Trip_ID, Driver_ID, Vehicle_ID, Trip_Date, Distance_KM, Avg_Speed_kmph, Max_Speed_kmph
Telemetry	One row per minute per trip (~12,987 rows)	Trip_ID, Timestamp, GPS, Speed_kmph, Accel_X/Y/Z_g, Gyro_X/Y/Z_dps

All relationships were configured as **Many-to-One** with a **single-direction filter** flowing from the Trips/Telemetry tables toward the master tables (Drivers and Vehicles).

3. Data Preparation & Cleaning

Before any metric was built, the following data preparation steps were carried out inside **Power Query Editor**:

- Removed blank and header rows that were imported from the Excel sheets.
- Corrected data types — several sensor columns (**Accel_X_g**, **Gyro_X_dps**, **License_Experience_Years**, **Odometer_KM_Start_of_Week**) were loaded as **Text** instead of Decimal/Whole Number. These were changed to their appropriate numeric types.
- Set **Data Category** for **Latitude** and **Longitude** columns in the Telemetry table to enable the GPS Map visual.

All event-detection logic (harsh braking, harsh acceleration, sharp cornering, speeding) was implemented as **Calculated Columns** directly in the Telemetry table. This was a deliberate choice — Calculated Columns evaluate row-by-row (once per telemetry minute), whereas

Measures evaluate across a filter context. Using Measures for binary flags would cause the "single value cannot be determined" error.

All KPIs, rates, and composite scores were then built as **Measures** in a dedicated `_Measures` table, keeping the data model clean and organised.

4. Dashboard 1 — Driver Behaviour Dashboard

4.1 Overview

This dashboard is designed to help fleet managers answer three questions at a glance:

- Who are the riskiest drivers in the fleet right now?
- What specific risky behavior is each driver exhibiting — speeding, harsh braking, harsh acceleration, or sharp cornering?
- Is the fleet getting riskier as the week progresses (a sign of driver fatigue)?

4.2 Event Detection — Calculated Columns in Telemetry

The following flag columns were created in the **Telemetry** table. Each flag is either 0 (normal) or 1 (event detected) for every single minute of every trip.

Flag Column	DAX Logic	Threshold Justification
Harsh_Braking_Flag	<code>Accel_X_g < -0.4</code>	A deceleration of 0.4g or more indicates sudden braking. Urban two-wheelers in Bengaluru traffic rarely need to brake this sharply in a controlled manner. Values beyond this consistently indicate emergency or aggressive braking behavior.
Harsh_Accel_Flag	<code>Accel_X_g > 0.4</code>	Same logic applied to the positive X-axis direction. An acceleration beyond 0.4g for a scooter/motorcycle in city

Flag Column	DAX Logic	Threshold Justification
		conditions points to deliberate aggressive throttle behavior.
Sharp_Corner_Flag	$ABS(Gyro_Z_dps) > 5$	The Z-axis gyroscope captures yaw — the rotation of the vehicle while turning. A reading of more than 5 degrees per second indicates a sharp, high-speed corner rather than a controlled turn.
Speeding_Flag	$Speed_kmph > 60$	60 km/h is the standard urban speed limit for two-wheelers in Bengaluru. Any minute where the vehicle's speed exceeds this threshold is counted as a speeding minute.

4.3 Exposure-Normalized Event Rates (Per 100 KM)

The initial approach used events **per trip** as the rate for scoring. This was identified as a calibration error — if one driver takes consistently short trips and another takes longer routes, comparing raw per-trip counts is misleading. A driver who covers more distance naturally has more opportunities to encounter situations requiring braking or acceleration.

The industry-standard fix is to normalize by distance: **events per 100 km**. This is the same metric used by commercial telematics providers (e.g., Modus, Lytx, Zubie).

$Harsh_Brake_Per_100km = (SUM\ of\ Harsh_Braking_Flag / Total\ Distance\ KM) \times 100$

$Harsh_Accel_Per_100km = (SUM\ of\ Harsh_Accel_Flag / Total\ Distance\ KM) \times 100$

$Harsh_Corner_Per_100km = (SUM\ of\ Sharp_Corner_Flag / Total\ Distance\ KM) \times 100$

$Overspeed_Pct = (SUM\ of\ Speeding_Flag / Total\ Telemetry\ Minutes) \times 100$

4.4 Driver Risk Score — Methodology

The Problem with Fixed Weights

The first version of the Driver Risk Score used hardcoded multipliers (e.g., `Speeding_Pct × 3`, `Harsh_Braking × 10`) to deduct points from 100. In practice, since the fleet's average harsh braking rate was very low (0.3–0.8 events per trip), even the riskiest driver was only losing 3–8 points. Every driver ended up with a score between 90–99, which made the ranking completely useless.

The Fix — Min-Max Normalization (Fleet-Relative Scoring)

Instead of scoring against an arbitrary hardcoded ceiling, each metric is normalized against the **actual minimum and maximum values observed within the fleet**. This guarantees that the riskiest driver always scores 100 on that dimension and the safest always scores 0, with everyone else spread proportionally in between.

$$\text{Norm_Score} = ((\text{Current Value} - \text{Fleet Minimum}) / (\text{Fleet Maximum} - \text{Fleet Minimum})) \times 100$$

This was implemented using `MINX(ALL(Drivers), [Metric])` and `MAXX(ALL(Drivers), [Metric])` in DAX, which iterates across all 30 drivers and returns the fleet-wide minimum and maximum, regardless of any slicer or filter applied on the canvas.

Assumption: Min-Max normalization is sensitive to extreme outliers. If one driver has an unusually extreme event count, it compresses all other drivers' scores toward the lower end. This trade-off was accepted for this one-week snapshot dataset. For a production system, **percentile rank scaling (RANKX-based)** would be more robust and is noted as a recommended improvement.

Final Composite Score (0 = Safest, 100 = Riskiest)

The four normalized component scores are combined using the following weights:

Component	Weight	Justification
Overspeed %	30%	Speeding is the single biggest contributing factor to road fatalities in India (MoRTH 2023 report). It receives the highest weight.
Harsh Braking per 100km	25%	Indicates consistently unpredictable riding — either

Component	Weight	Justification
		tailgating or poor situational awareness. Weighted equally with cornering.
Sharp Cornering per 100km	25%	High-speed cornering on a two-wheeler is a leading cause of side-slip crashes. Equally weighted with braking.
Harsh Acceleration per 100km	20%	Indicates aggressive riding but has slightly lower crash risk than braking and cornering events. Weighted lowest.

Driver_Risk_Score =

$(0.30 \times \text{Norm_Overspeed_Risk}) +$

$(0.25 \times \text{Norm_Brake_Risk}) +$

$(0.20 \times \text{Norm_Accel_Risk}) +$

$(0.25 \times \text{Norm_Corner_Risk})$

Risk Tier Classification — Percentile-Based

Rather than using hardcoded thresholds (e.g., score > 70 = "High Risk"), the Risk Tier is assigned dynamically using **fleet percentiles**. This ensures that the tier labels always reflect the current distribution of 30 drivers and do not become meaningless if the entire fleet improves or worsens over time.

P66 = 66th Percentile of Driver_Risk_Score across ALL Drivers

P33 = 33rd Percentile of Driver_Risk_Score across ALL Drivers

High Risk → Score $\geq$ P66 (top third of the fleet by risk)

Medium Risk → Score $\geq$ P33 (middle third)

Low Risk → Score < P33 (bottom third — safest drivers)

4.5 Visuals — Dashboard 1

KPI Cards (Top Row)

Five headline figures are displayed at the top of the dashboard:

Card	Value	Interpretation
Total Distance	6.15K km	Total fleet distance covered in the observed week.
Total Trips	450	30 drivers × 15 trips each, confirming complete data coverage.
Avg Fleet Risk Score	28.19	The average risk score across all 30 drivers. By definition (min-max normalization), this will always sit around the midpoint.
High Risk Drivers	24	The count of drivers in the top risk tier (above P66).
Total Drivers	30	Total active drivers in the fleet.

Visual 1 — Driver Risk Ranking (Horizontal Bar Chart)

Drivers are sorted in ascending order of their Risk Score so the riskiest drivers appear at the top. Bars are conditionally formatted Red (High Risk), Yellow (Medium Risk), and Green (Low Risk) using the `Vehicle_Health_Flag` rules logic. A dashed reference line is placed at the P66 threshold.

Visual 2 — Harsh Events Breakdown (Stacked Bar Chart)

Shows the proportion of each event type (Harsh Braking, Harsh Acceleration, Sharp Cornering) as a stacked bar per driver. This visual cross-filters with the Risk Ranking — clicking a driver in the ranking immediately highlights their specific behavioral breakdown. This is where fleet managers identify whether to send a driver for braking-specific training vs. speed counseling.

Visual 3 — Driver Aggressiveness Matrix (Scatter Plot)

- **X-axis:** Average Speed (kmph)
- **Y-axis:** Harsh Events per 100 km
- **Bubble Size:** Total Distance KM
- **Color:** Experience Group (Novice / Intermediate / Veteran)

Median reference lines divide the scatter into four quadrants. Drivers in the **top-right quadrant** (high speed AND high harsh events) are the most dangerous. Drivers in the **bottom-left quadrant** (low speed AND low events) are the safest. The bubble size reveals whether a driver's aggression comes from high mileage exposure or genuine risky behavior in a short distance.

Visual 4 — Daily Fleet Risk Trend (Line Chart)

- **X-axis:** Trip Date (7 days)
- **Y-axis:** Avg Fleet Risk Score

A trend line is overlaid on this chart. An upward slope across the week is a signal of **driver fatigue** — a known phenomenon in delivery operations where risk behavior increases toward the end of a work week. This chart directly answers the "late-week drift" hypothesis raised in the problem brief.

Visual 5 — GPS Route & Speed Map

Telemetry GPS coordinates are plotted on a map centered on Bengaluru. Bubble size is set to the average **Speed_kmph**. This creates a visual density map of fleet routes and speed hotspots across the city. When a single driver is selected via a slicer, the map automatically zooms to their specific route for that week.

Slicers

Four slicers are arranged in a right-side filter pane:

1. **Trip Date Range** (Between date slider)
2. **Home Hub** (List — Bellandur, HSR Layout, Rajajinagar, etc.)
3. **Experience Group** (Tile — Novice / Intermediate / Veteran)
4. **Gender** (Tile — Male / Female)

5. Dashboard 2 — Vehicle Health Status Dashboard

5.1 Overview

This dashboard answers a different set of questions:

- Which vehicles are showing the highest mechanical vibration (a proxy for suspension or engine wear)?
- Which vehicles are overdue for servicing?
- Is the vibration of a vehicle correlated with its age or odometer reading (validating the scoring model)?
- Are any vehicles degrading mid-week (sudden vibration increase in later days)?

5.2 Health Metrics — Methodology

Vibration Index (STDEV of Accel_Z_g)

The Z-axis accelerometer captures vertical vibrations — up-down movement of the vehicle. On a two-wheeler, this directly reflects the state of the suspension system, road smoothness aside. A vehicle with degrading suspension or engine mounts will produce **higher and more variable** Z-axis readings across a trip.

The **Standard Deviation** of `Accel_Z_g` across all telemetry minutes for a given vehicle was chosen as the Vibration Index. STDEV was preferred over a simple Average because:

- A high average `Accel_Z` could simply reflect a bumpy road that day.
- A consistently **high standard deviation** indicates the vehicle is producing irregular, unpredictable vibration — which is the actual signal of mechanical wear.

`Vibration_Index_Z = STDEV.S(Telemetry[Accel_Z_g])`

Gyro Noise Index

Wheel wobble, loose bearings, or chassis instability manifest as irregular rotational readings on the gyroscope axes. The Gyro Noise Index is the sum of the standard deviations of all three gyro axes, providing a single number that captures overall rotational instability.

`Gyro_Noise_Index = STDEV.S(Gyro_X_dps) + STDEV.S(Gyro_Y_dps) + STDEV.S(Gyro_Z_dps)`

Days Since Last Service

Since the dataset represents a snapshot from August 2026, `TODAY()` was replaced with `MAX(Trips[Trip_Date])` to calculate service gaps accurately relative to the dataset's own timeline.

```
Days_Since_Last_Service = DATEDIFF(MAX(Vehicles[Last_Service_Date]),  
MAX(Trips[Trip_Date]), DAY)
```

Vehicle Health Flag

The Health Flag combines both the vibration dimension and the service recency dimension into a single categorical label. Thresholds were calibrated against the actual fleet data distribution (max `Vibration_Index_Z` observed = 0.16, Avg Days Since Service = 32.7 days).

Flag	Vibration Condition	Service Condition	Interpretation
Service Needed	> 0.11	OR > 50 days	Active mechanical concern or significantly overdue service
Watch	> 0.08	OR > 30 days	Approaching thresholds — schedule proactively
Healthy	Below both	Below both	No immediate action required

Assumption: The vibration thresholds (0.08 and 0.11) were determined by examining the fleet's actual `Vibration_Index_Z` distribution (range: 0.03 – 0.16) rather than using a generic industry standard. In a production environment, a rolling 30-day baseline per vehicle model/make would be a more accurate reference point.

5.3 Visuals — Dashboard 2

KPI Cards (Top Row)

Card	Value	Interpretation
Avg Days Since Service	32.70	The fleet-wide average. Indicates most vehicles are due for service soon.
Total Vehicles	30	Full fleet coverage confirmed.
Vehicles Flagged for Service	7	7 out of 30 vehicles (23.3%) currently require attention.
Avg Fleet Vibration	0.07	Fleet-level average Vibration Index (STDEV of Accel_Z).
Total Fleet KM	6.15K	Total distance driven across the week (same figure as Dashboard 1).

Visual 1 — Vehicle Vibration Ranking (Horizontal Bar Chart)

Vehicles are sorted by their **Vibration_Index_Z** in descending order — the most vibrating (and therefore highest-risk) vehicles appear at the top. Bar colors are conditionally formatted using the same Red/Yellow/Green scheme as the Health Flag: Bars above the 0.11 threshold are Red, between 0.08 and 0.11 are Yellow, and below 0.08 are Green.

Visual 2 — Fleet Health Mix (Donut Chart)

A proportional view of the fleet split across the three health categories. In the current dataset: 56.67% Healthy, 23.33% Watch, 20% Service Needed. This gives management an instant risk-to-healthy ratio for the fleet without having to look through individual vehicle data.

Visual 3 — Wear Correlation Scatter Plot (Data Science Validation)

- **X-axis:** Odometer KM Start of Week (Maximum per vehicle)
- **Y-axis:** Vibration_Index_Z
- **Color:** Vehicle_Health_Flag

This is the most analytically significant visual on Dashboard 2. Its purpose is to **validate the scoring model** — if the Health Flag is correctly identifying mechanical wear, then vehicles with higher odometer readings should cluster at the top of the Y-axis (higher vibration). A positive

correlation (upward-trending scatter) confirms that the vibration index is genuinely capturing physical wear rather than random sensor noise. A trendline is added via the Analytics pane.

Visual 4 — Service Overdue (Column Chart with Threshold Line)

Each vehicle's `Days_Since_Last_Service` is shown as a vertical bar. A constant red dashed line is drawn at 60 days (the defined "Service Needed" threshold). Any bar crossing this line is immediately visible. Bars are conditionally formatted Red, Yellow, or Green using the same Health Flag rules.

Visual 5 — Daily Vibration Degradation (Line Chart with Small Multiples)

- **X-axis:** Trip Date
- **Y-axis:** Vibration_Index_Z
- **Small Multiples:** Vehicle_ID

Plotting 30 vehicles on a single line chart would create an unreadable spaghetti chart. Using Small Multiples creates a grid of mini line-charts — one per vehicle. This allows a fleet manager to spot a specific vehicle whose vibration starts low on Monday but increases sharply by Thursday. That kind of within-week degradation pattern is a strong indicator of a developing mechanical fault, not just a noisy road.

Slicers

Three slicers are available in the right-side filter panel:

1. **Make** (Tile — Bajaj, Hero, Honda, Ola, Suzuki, TVS, Yamaha)
2. **Model** (Dropdown — Aactiva, Ntorq, Pulsar, Ray ZR, etc.)
3. **Manufacture Year** (Dropdown — 2019 to 2025)

6. Cross-Dashboard Interaction

Both dashboards share the same underlying data model. The `_Measures` table holds all calculated measures centrally, and the same `Total_Distance_KM` and `Total_Fleet_KM` figures appear on both dashboards — ensuring consistency.

When opened together, a fleet manager can:

1. Identify a **High Risk Driver** on Dashboard 1.
2. Look up their **Primary Vehicle ID** in the driver telematics table.
3. Switch to Dashboard 2 and filter by that Vehicle ID to check whether the vehicle itself is also showing wear — separating driver behavior risk from vehicle mechanical risk.

7. What Else Could This Dataset Be Used For?

Beyond the two dashboards built for this assignment, this dataset has clear potential for the following applications:

Use Case	How
Route Optimization	Cluster GPS coordinates using K-Means to identify inefficient or high-risk route corridors across Bengaluru hubs.
Shift & Schedule Planning	The daily risk trend data can be extended to optimize shift timings — avoiding assigning back-to-back high-distance days to drivers who show fatigue-related risk increase.
Insurance Risk Profiling	The Driver Risk Score is directly usable as an input to a usage-based insurance (UBI) premium model. Higher-risk drivers are assigned higher premiums or required to complete safety training before eligibility for coverage.
Predictive Maintenance Model	Train a binary classification model (Logistic Regression or XGBoost) on historical vibration trends to predict vehicle breakdown probability 7–14 days in advance — rather than relying on threshold rules.
Driver Training Prioritization	The event breakdown visual already identifies whether a driver's risk is driven by speeding, braking, or cornering. This can be directly fed into a personalized training program where each driver receives targeted coaching on their specific weak area.
Fuel & Battery Efficiency Analysis	Harsh acceleration and overspeed events directly correlate with excess fuel consumption (for petrol vehicles) and battery drain (for Ola S1 EVs). This dataset can quantify per-driver fuel cost variance across the fleet.

